

Homework: Cost Management Report IT Project Management

Exercises 1 and 2

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1 Exercise 1: Mobile App Development Project

1.1 Given Data

- Budget at Completion (BAC): \$200,000
- Planned Value (PV): \$80,000
- Earned Value (EV): \$60,000
- Actual Cost (AC): \$75,000

1.2 Calculations

1. Cost Performance:

- Cost Variance: $CV = EV - AC = 60,000 - 75,000 = -15,000$
- Cost Performance Index: $CPI = \frac{EV}{AC} = \frac{60,000}{75,000} = 0.8$

2. Schedule Performance:

- Schedule Variance: $SV = EV - PV = 60,000 - 80,000 = -20,000$
- Schedule Performance Index: $SPI = \frac{EV}{PV} = \frac{60,000}{80,000} = 0.75$

3. To-Complete Performance Index (TCPI):

- $TCPI = \frac{BAC - EV}{BAC - AC} = \frac{200,000 - 60,000}{200,000 - 75,000} = \frac{140,000}{125,000} = 1.12$

1.3 Conclusion

- **Budget Status:** The project is **over budget** since $CV < 0$ and $CPI < 1$.
- **Schedule Status:** The project is **behind schedule** since $SV < 0$ and $SPI < 1$.
- **Recovery Ability:** With a $TCPI$ of 1.12, completing the project within the original budget is **challenging but possible** with improved efficiency.

2 Exercise 2: 1-Year IT Project

2.1 Given Data

- Budget at Completion (BAC): \$120,000
- Time at Completion (TAC): 12 months
- Planned Value (PV): \$23,000
- Earned Value (EV): \$20,000
- Actual Cost (AC): \$25,000

2.2 Calculations

1. Cost Status:

- $CV = EV - AC = 20,000 - 25,000 = -5,000$
- $CPI = \frac{EV}{AC} = \frac{20,000}{25,000} = 0.8$
- **Result:** The project is currently **over budget**.

2. Schedule Status:

- $SV = EV - PV = 20,000 - 23,000 = -3,000$
- $SPI = \frac{EV}{PV} = \frac{20,000}{23,000} \approx 0.87$
- **Result:** The project is **behind schedule**.

3. Estimate At Completion (EAC):

- $EAC = \frac{BAC}{CPI} = \frac{120,000}{0.8} = 150,000$
- **Result:** The total projected cost at completion is **\$150,000**.

4. Estimated Time to Complete (ETC - Time):

- Total Estimated Duration: $EAC_{time} = \frac{TAC}{SPI} = \frac{12}{0.87} \approx 13.79$ months
- Assuming the current time elapsed is proportional to the PV (≈ 2.3 months):
- $ETC_{time} = EAC_{time} - \text{Current Time} \approx 11.49$ months
- **Result:** The project is estimated to take approximately **13.79 months** in total.